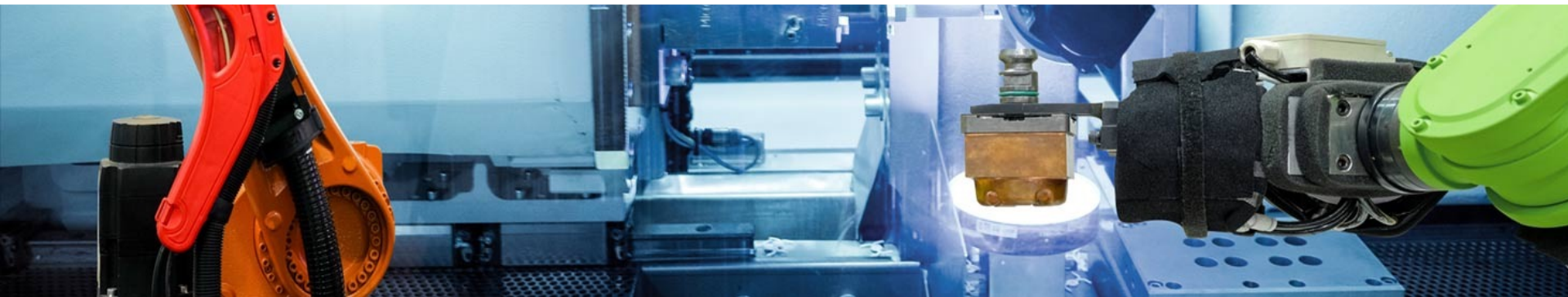




XU THẾ PHÁT TRIỂN CỦA HỆ THỐNG CƠ ĐIỆN TỬ TIỀN TIẾN



Contents



- Short Introduction of Mechatronic Systems
- Mechatronic System Design
- Recent Trends in Mechatronics
- Challenges of Mechatronic System

Short Introduction of Mechatronic Systems

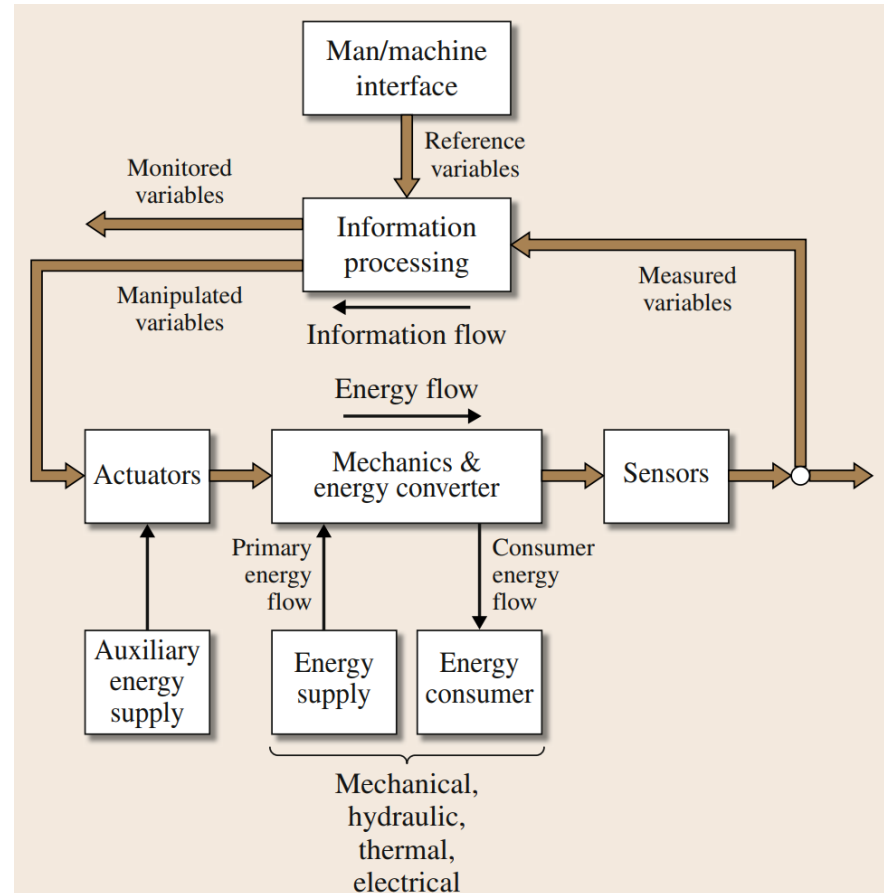
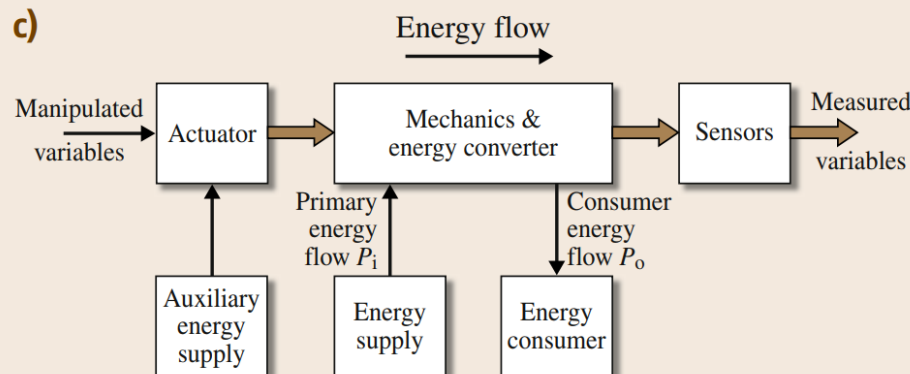
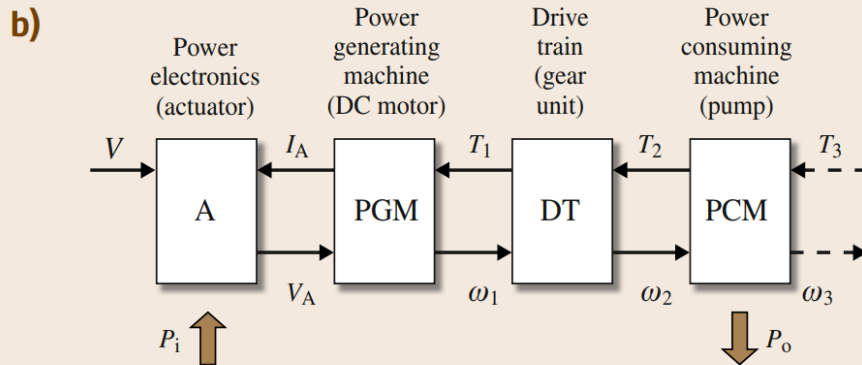
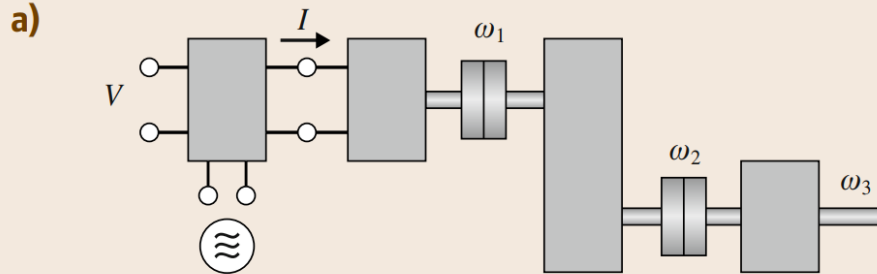
- From Mechanical to Mechatronic Systems
- Examples of Mechatronic Systems
- Functions of Mechatronic Systems



From Mechanical to Mechatronic Systems

Schematic representation of a machine set

V – voltage; V_A – armature voltage;
 I_A – armature current;
 T – torque;
 ω – angular frequency;
 P_i – drive power;
 P_o – consumer power

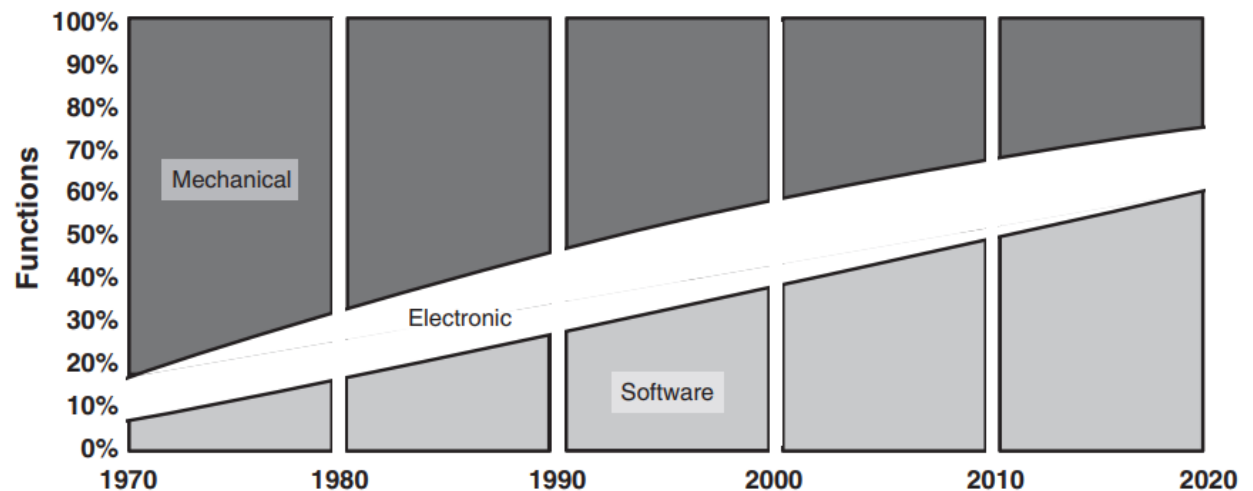
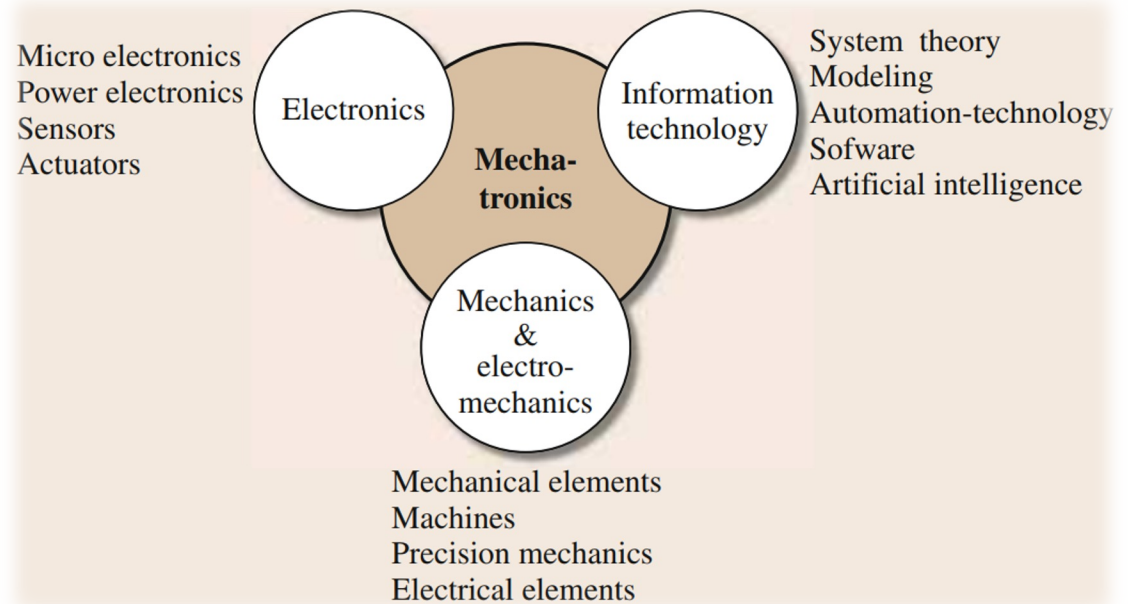


From Mechanical to Mechatronic Systems

Mechatronics: synergetic integration of different disciplines

These integrated mechanical–electronic systems are increasingly called **mechatronic systems**.

The word *mechatronics* was probably first created by a Japanese engineer, Tetsuro Mori, in 1969 and had a trademark by a Japanese company until 1972

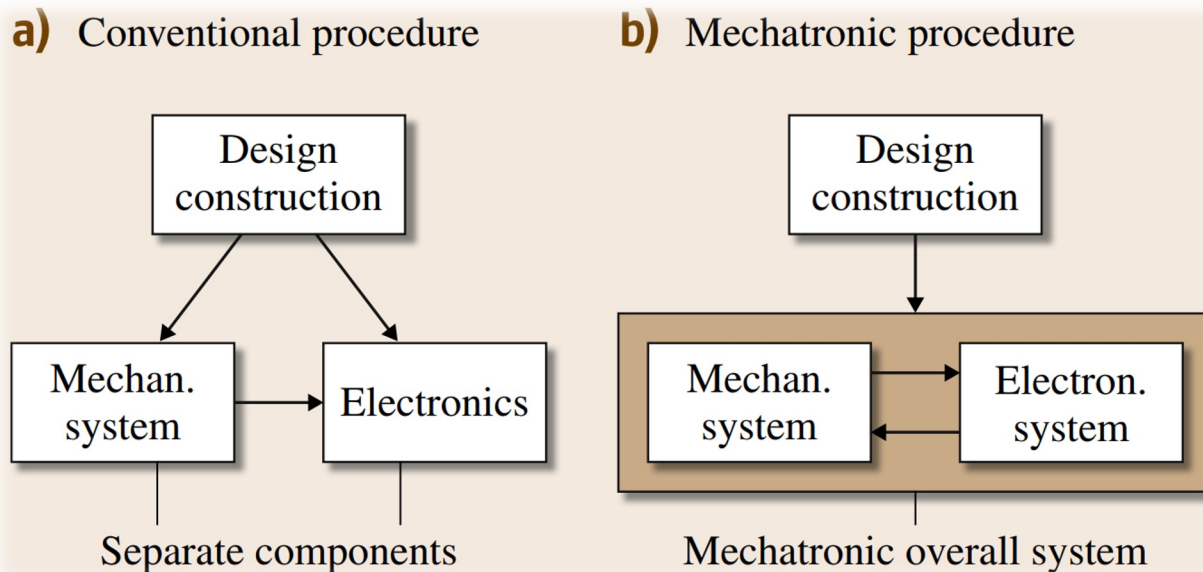


Difference of Industrial Automation and Mechatronics

While **mechatronics** and **industrial automation** are two fields that share certain similarities, they have many distinct differences as well.

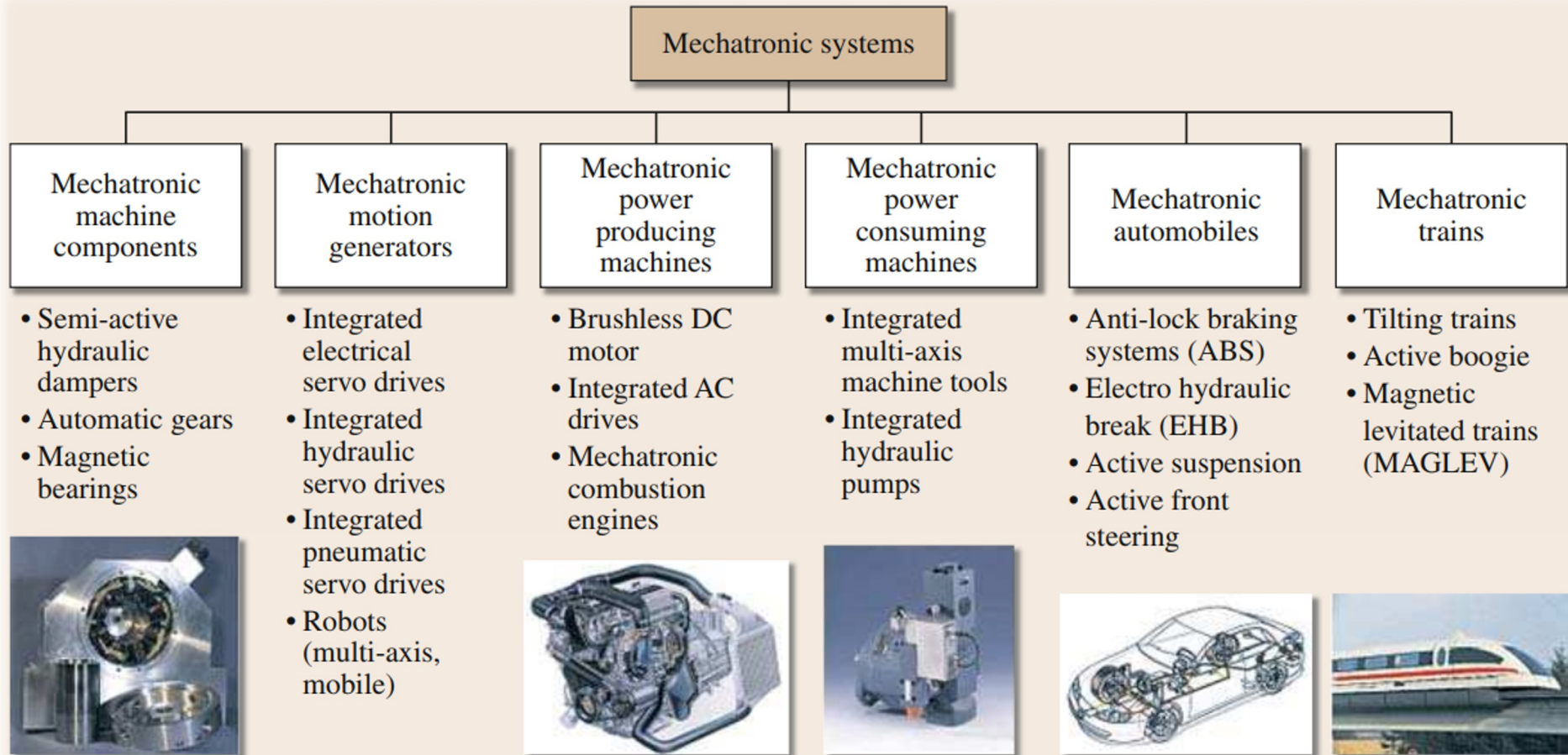
The solution of tasks to design mechatronic systems is performed on the mechanical as well as on the digital-electronic side.

➤ Interrelations during the design and construction of mechatronic systems.



Simultaneous engineering has to take place, with the goal of designing an overall integrated system (an *organic system*) and also creating synergetic effects

Examples of Mechatronic Systems



Functions of Mechatronic Systems

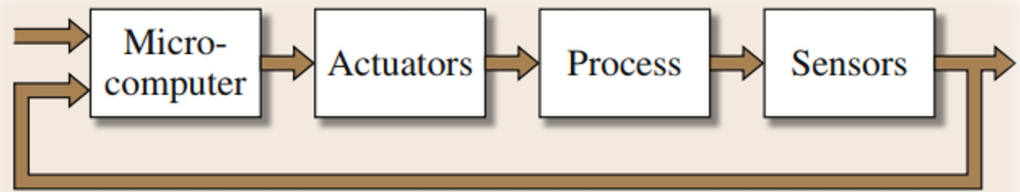
Properties of conventional and mechatronic designed systems

Conventional design	Mechatronic design
Added components	Integration of components (hardware)
Bulky	Compact
Complex	Simple mechanisms
Cable problems	Bus or wireless communication
Connected components	Autonomous units
Simple control	Integration by information processing (software)
Stiff construction	Elastic construction with damping by electronic feedback
Feedforward control, linear (analog) control	Programmable feedback (nonlinear) digital control
Precision through narrow tolerances	Precision through measurement and feedback control
Nonmeasurable quantities change arbitrarily	Control of nonmeasurable estimated quantities
Simple monitoring	Supervision with fault diagnosis
Fixed abilities	Adaptive and learning abilities

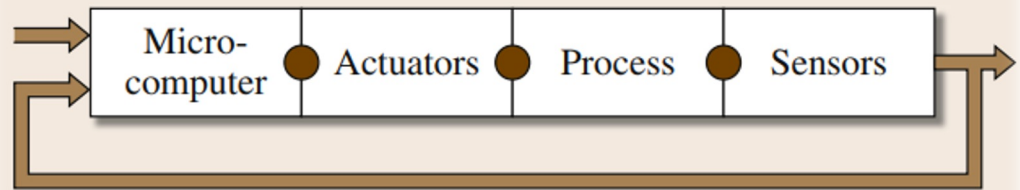
Integration Forms of Processes with Electronics

- General scheme of a (classical) mechanical–electronic system

Integration of mechatronic systems

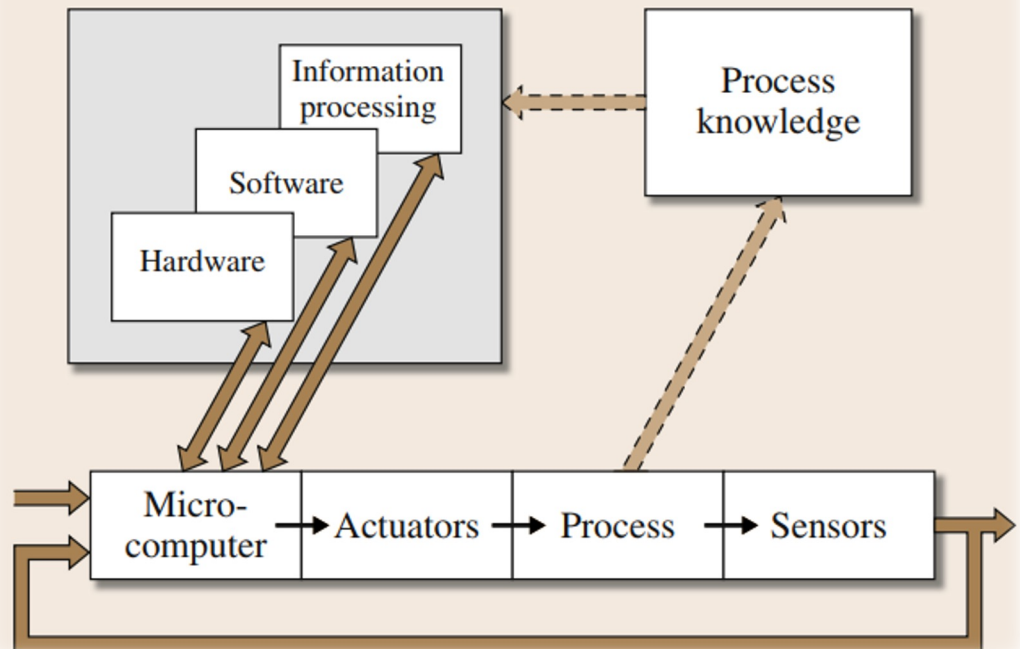


- Integration through components (hardware integration)



● Possible points of integration

- Integration through functions (software integration)



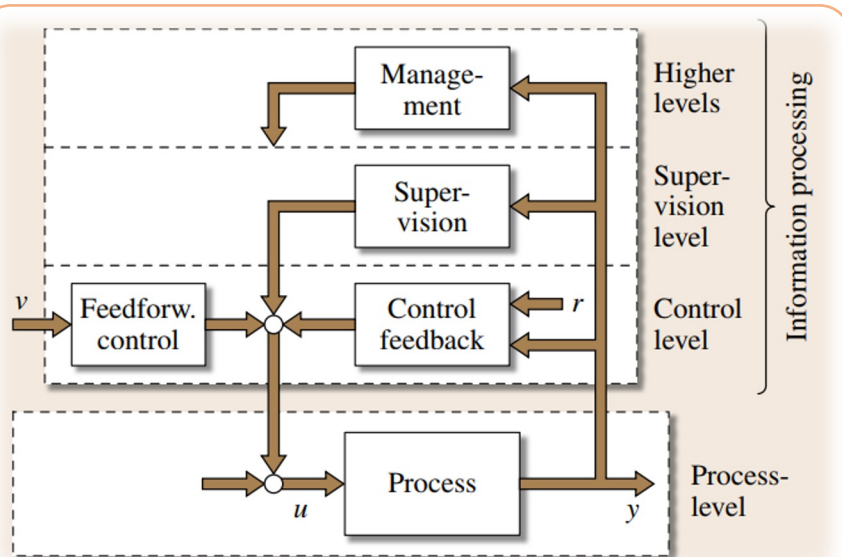
Integration Forms of Processes with Electronics

- Integration through components
(hardware integration)



Integration Forms of Processes with Electronics

- Integration by information processing (software integration)



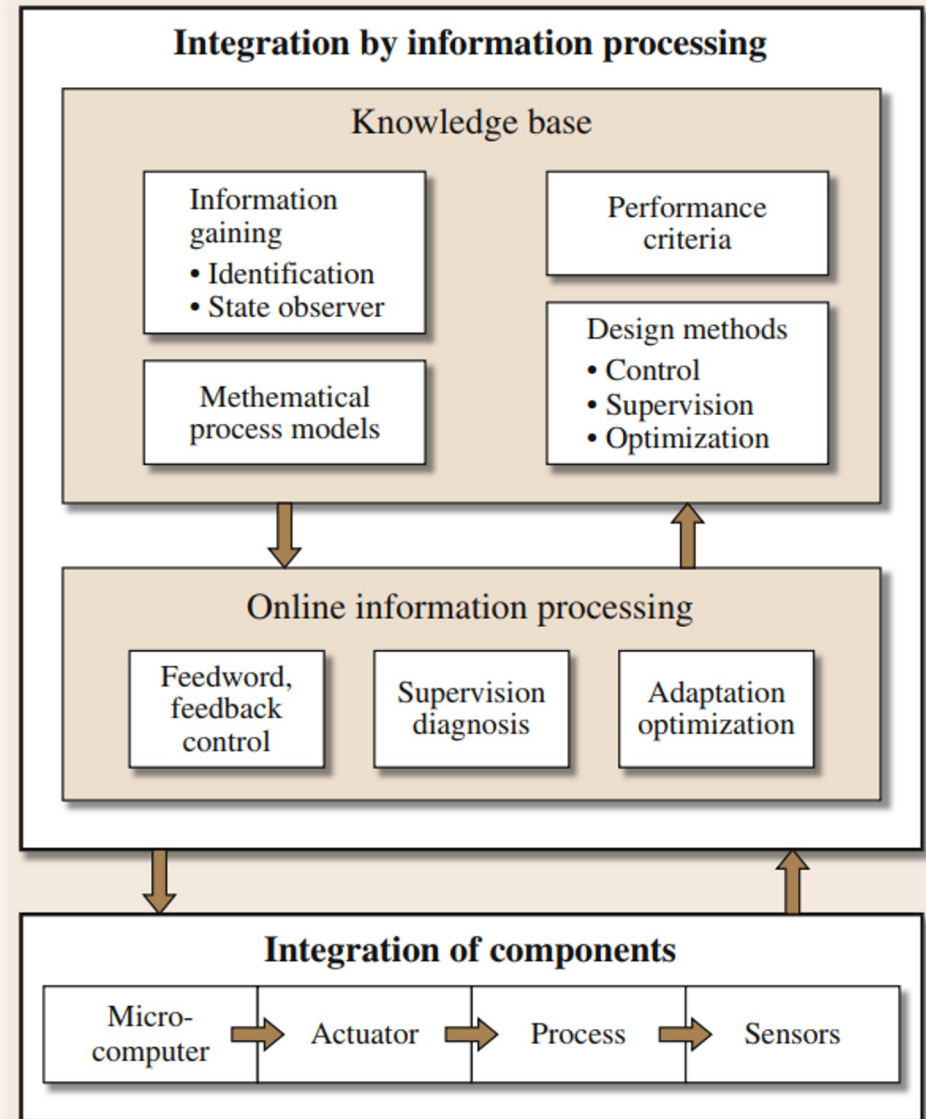
Different levels of information processing for process automation.

u : manipulated variables;

y : measured variables;

v : input variables;

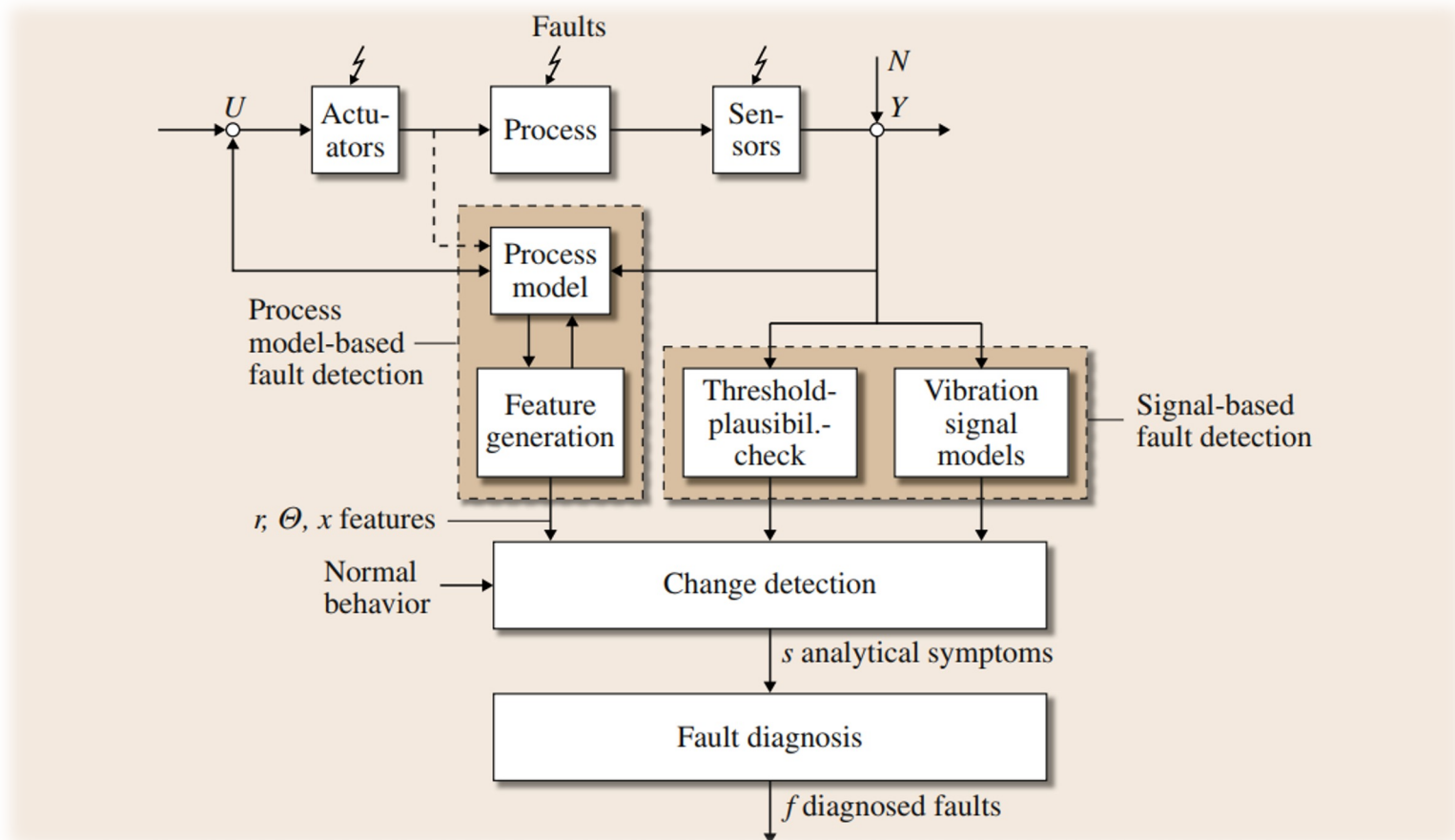
r : reference variables



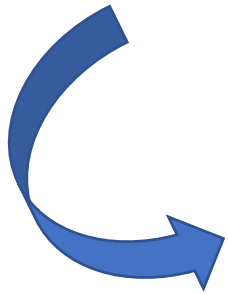
Integration Forms of Processes with Electronics

With the increasing number of automatic functions (autonomy) including electronic components, sensors, and actuators, increasing complexity, and increasing demands on reliability and safety, integrated supervision with fault diagnosis becomes increasingly important.

➤ Fault detection is a significant natural feature of an intelligent mechatronic system



Discussion



Mechatronic System Design

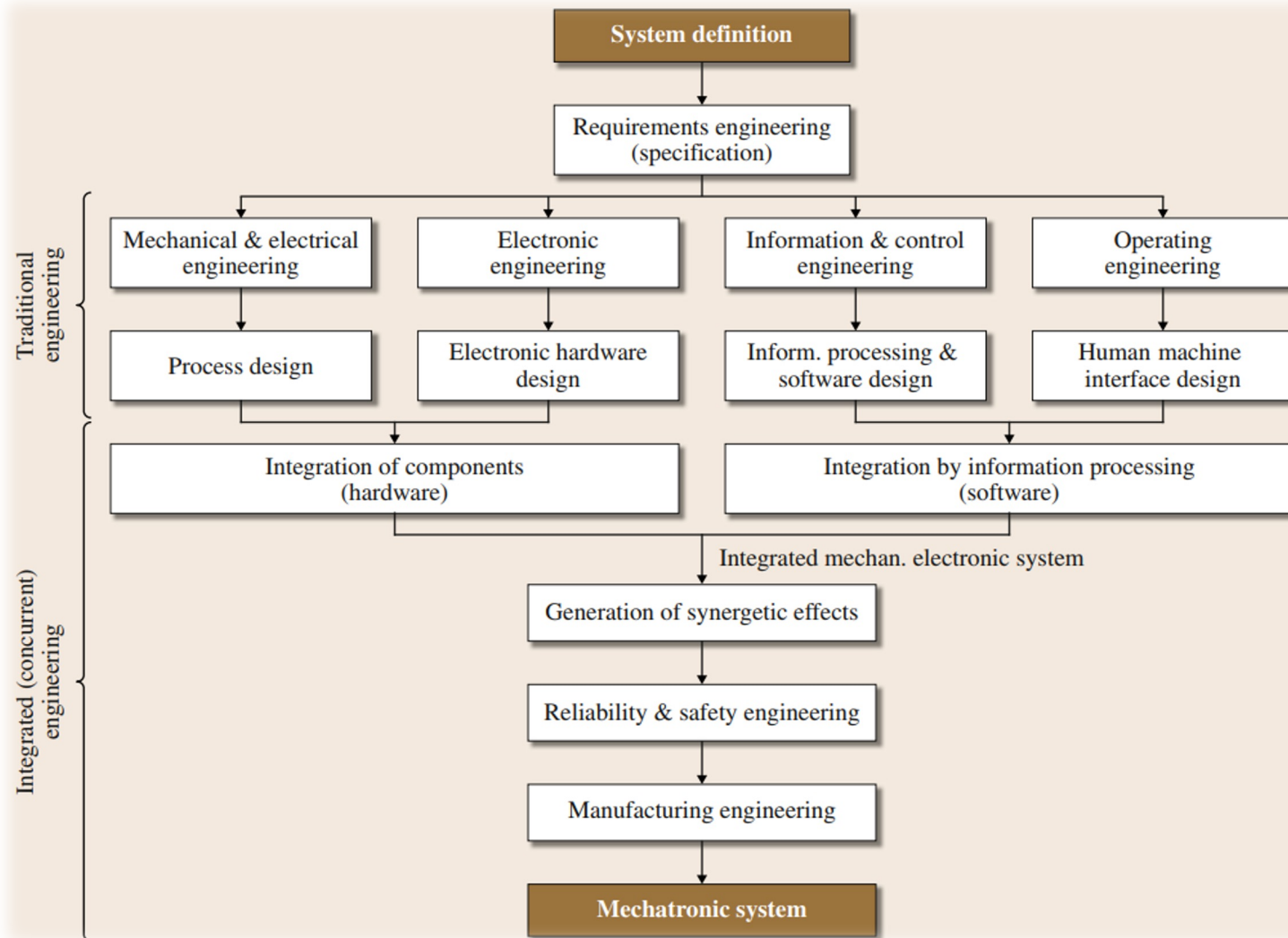
- Design Procedures for Mechatronic Systems
- Computer-Aided Design of Mechatronic Systems



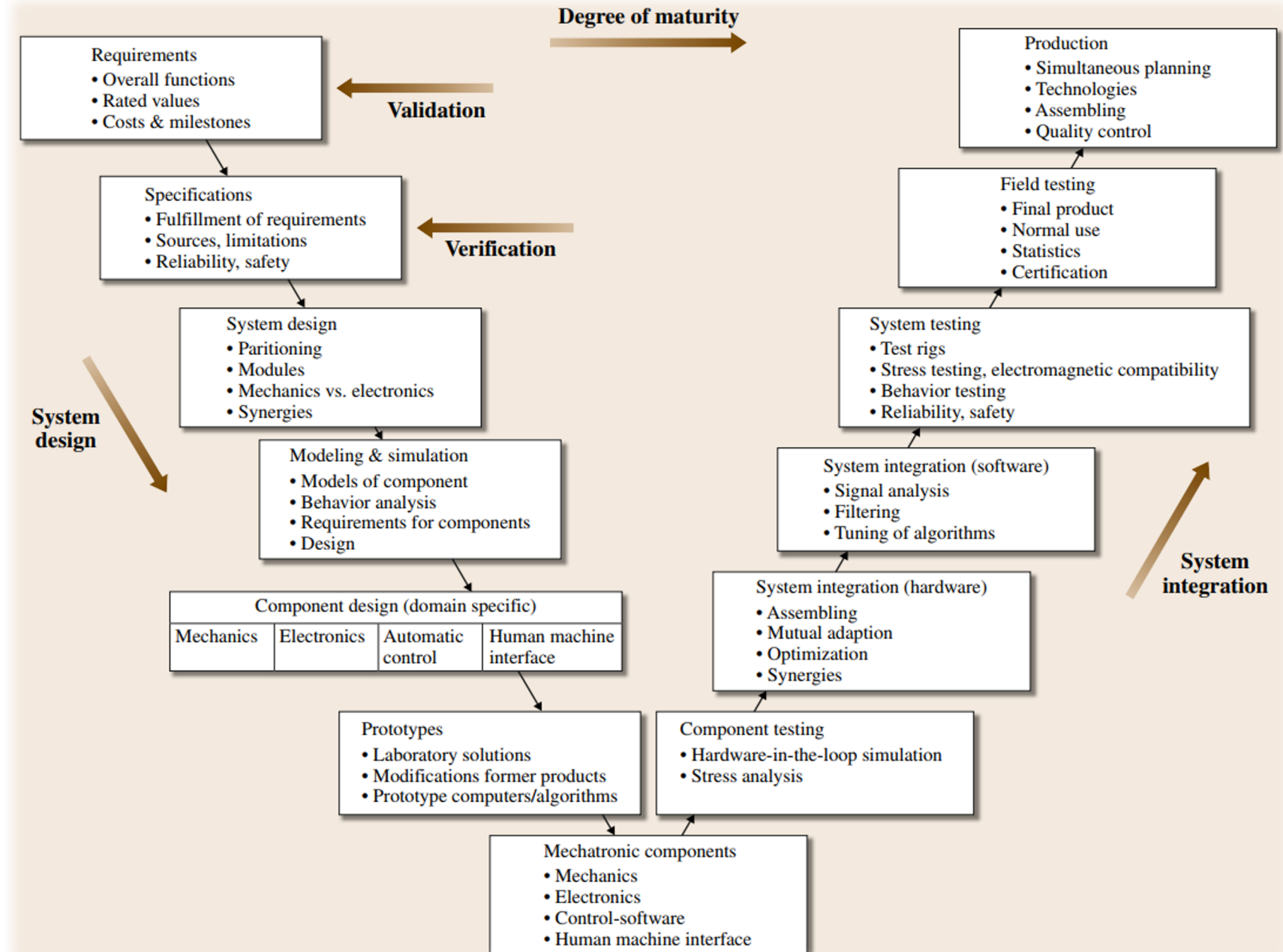
Design Procedures for Mechatronic Systems

The design of mechatronic systems requires systematic development and use of modern software design tools.

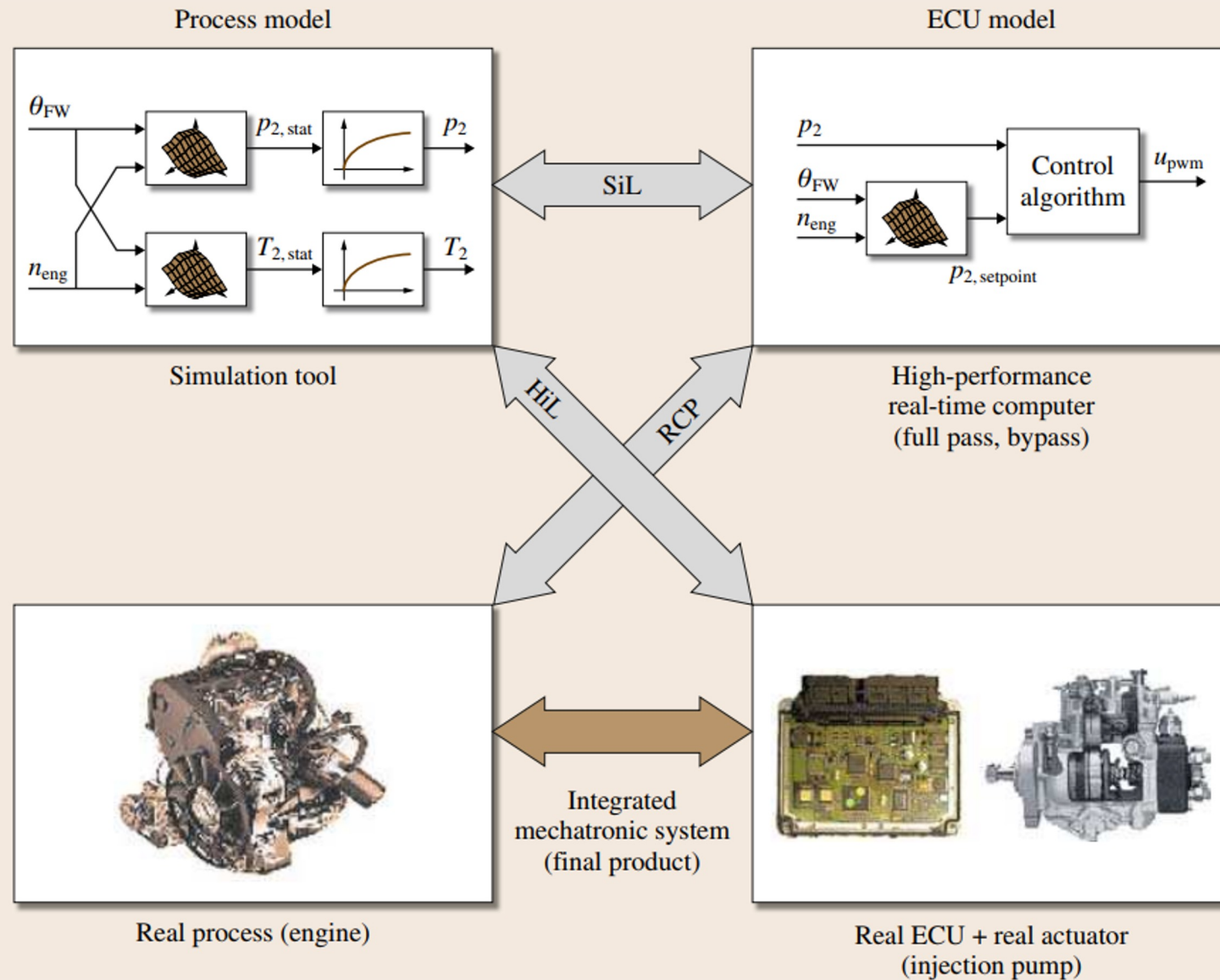
➤ Mechatronic design is also an iterative procedure.



“V” development scheme for mechatronic systems



Computer-Aided Design of Mechatronic Systems

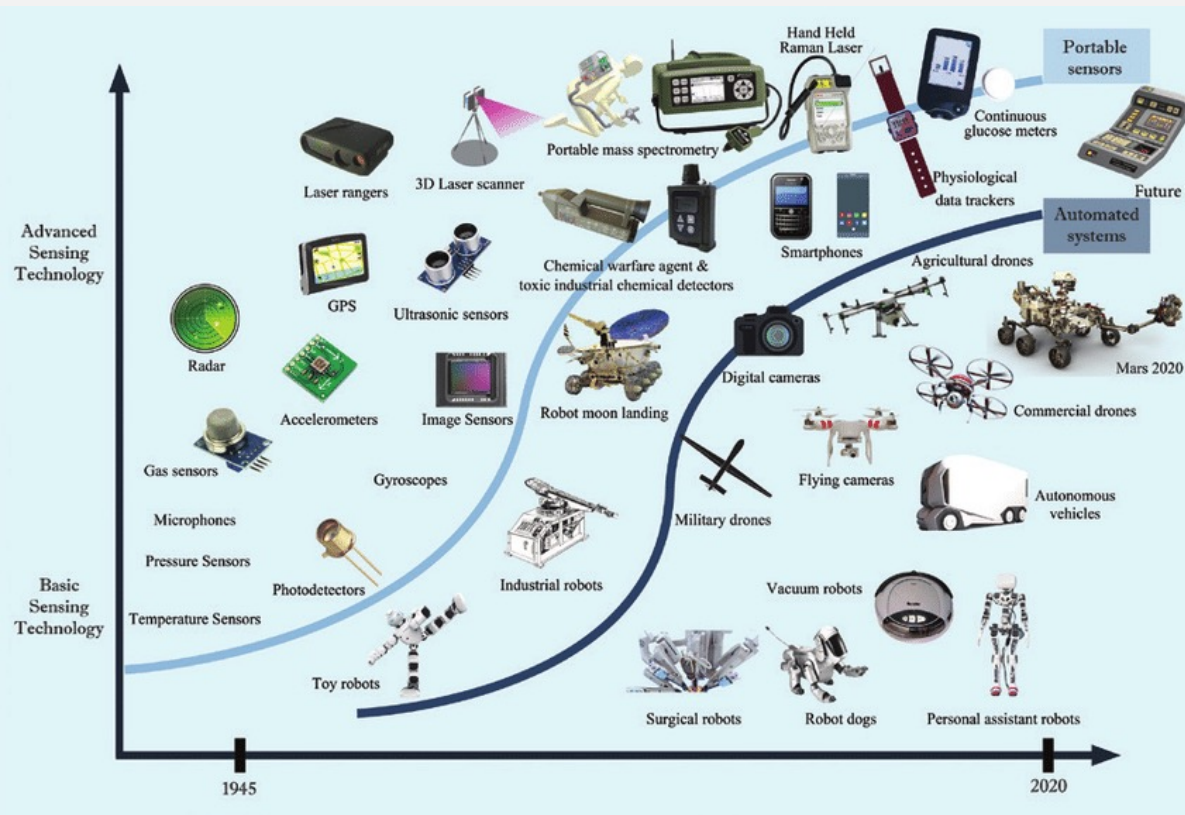


Recent Trends in Mechatronics

- Advanced Sensors and Actuators
- Internet of Things (IoT)
- Cyber-Physical Systems (CPS), Digital Twin, and Human-Machine Interface (HMI)
- Artificial Intelligence (AI) and Machine Learning (ML)

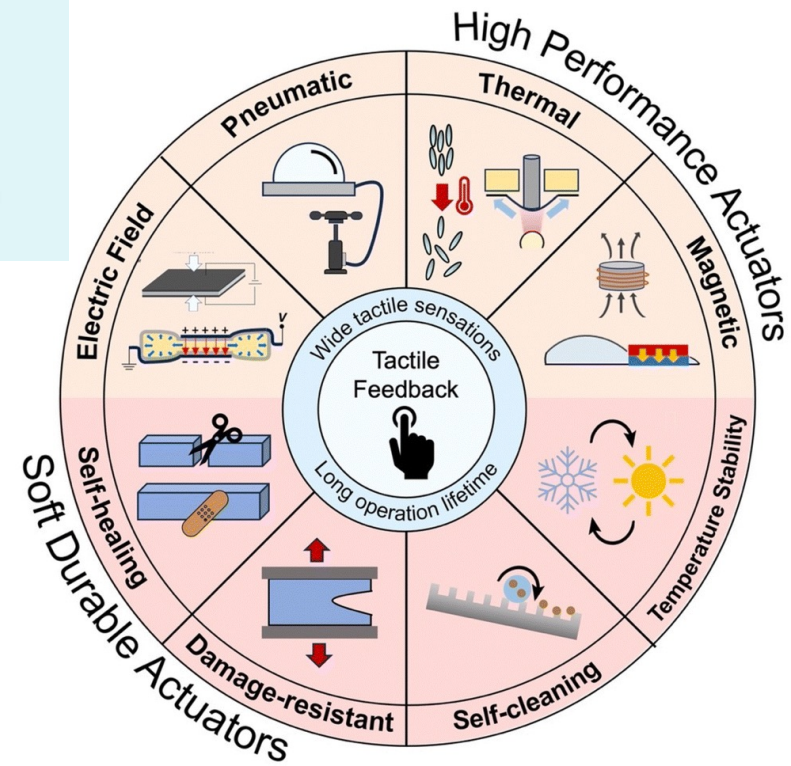


Advanced Sensors and Actuators

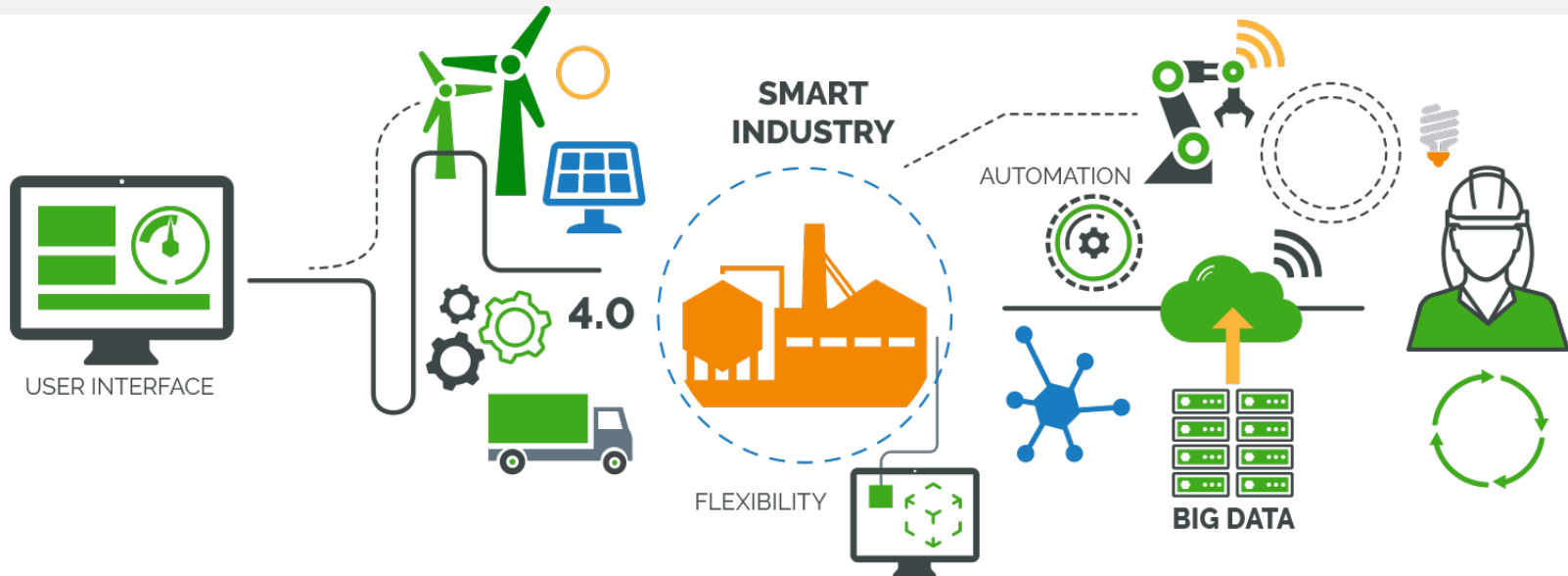


- Micro-Electro-Mechanical Systems (MEMS)
- Tactile/Touch Sensors
- Flexible/Stretchable Sensors
- Wearable Health Sensors
- Soft Actuators
- Piezoelectric Actuators
- Bio-Inspired Actuators

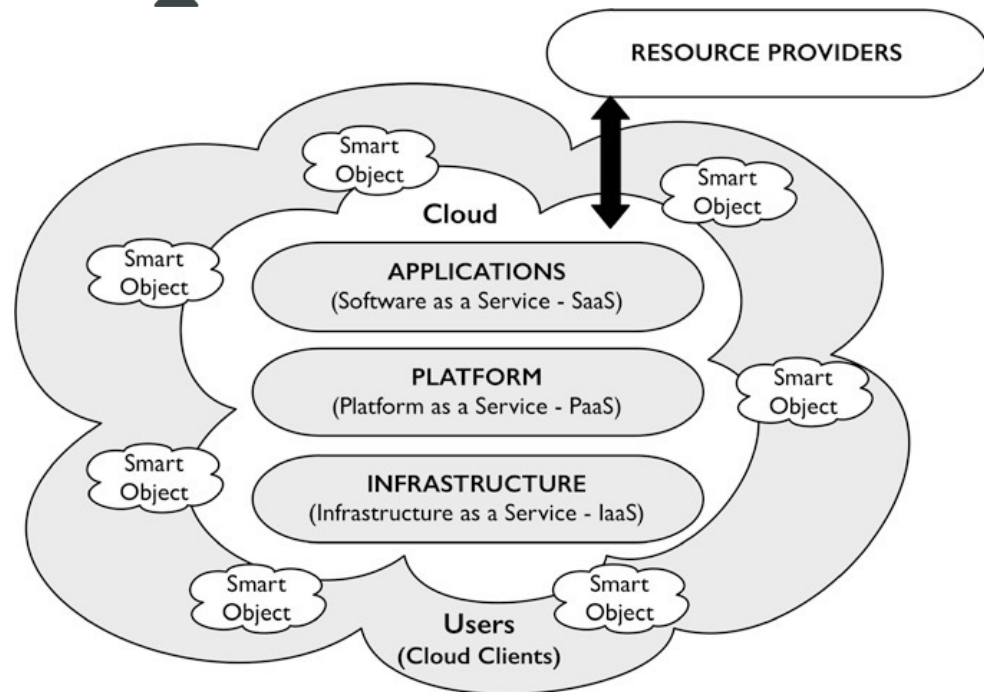
Advanced Sensors and Actuators: These are key components in modern mechatronic systems, enabling precise measurement and control. Advanced sensors gather accurate real-time data, while actuators convert this data into actions, enhancing system efficiency and performance across industries.



Cloud-based structures for the Internet of Things



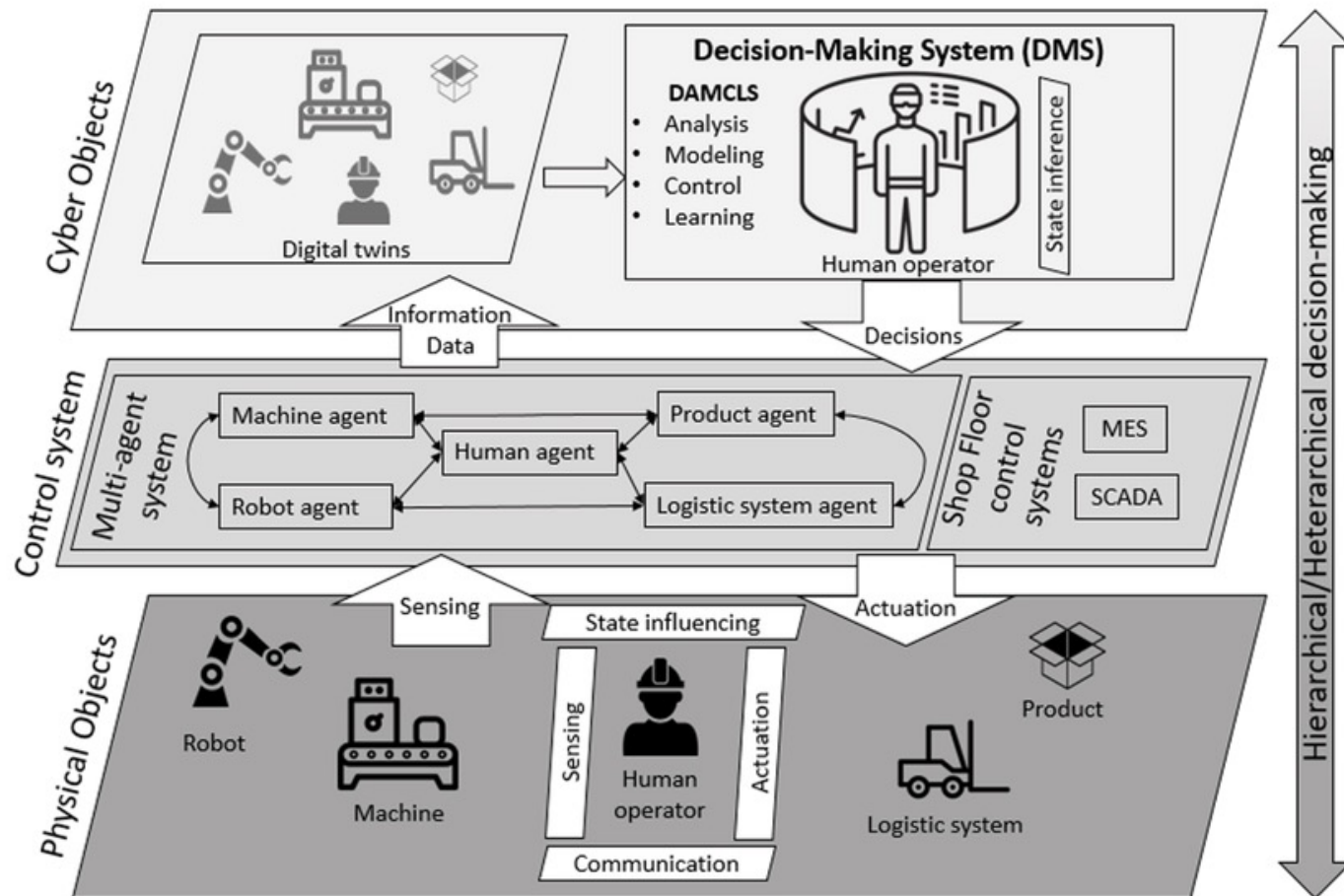
Cloud-based structures for the Internet of Things (IoT): These are scalable, flexible platforms that allow IoT devices to collect, store, and process data in real-time via cloud computing. By leveraging the cloud, IoT systems can achieve enhanced connectivity, data analytics, and remote control, enabling smarter and more efficient applications across various industries.



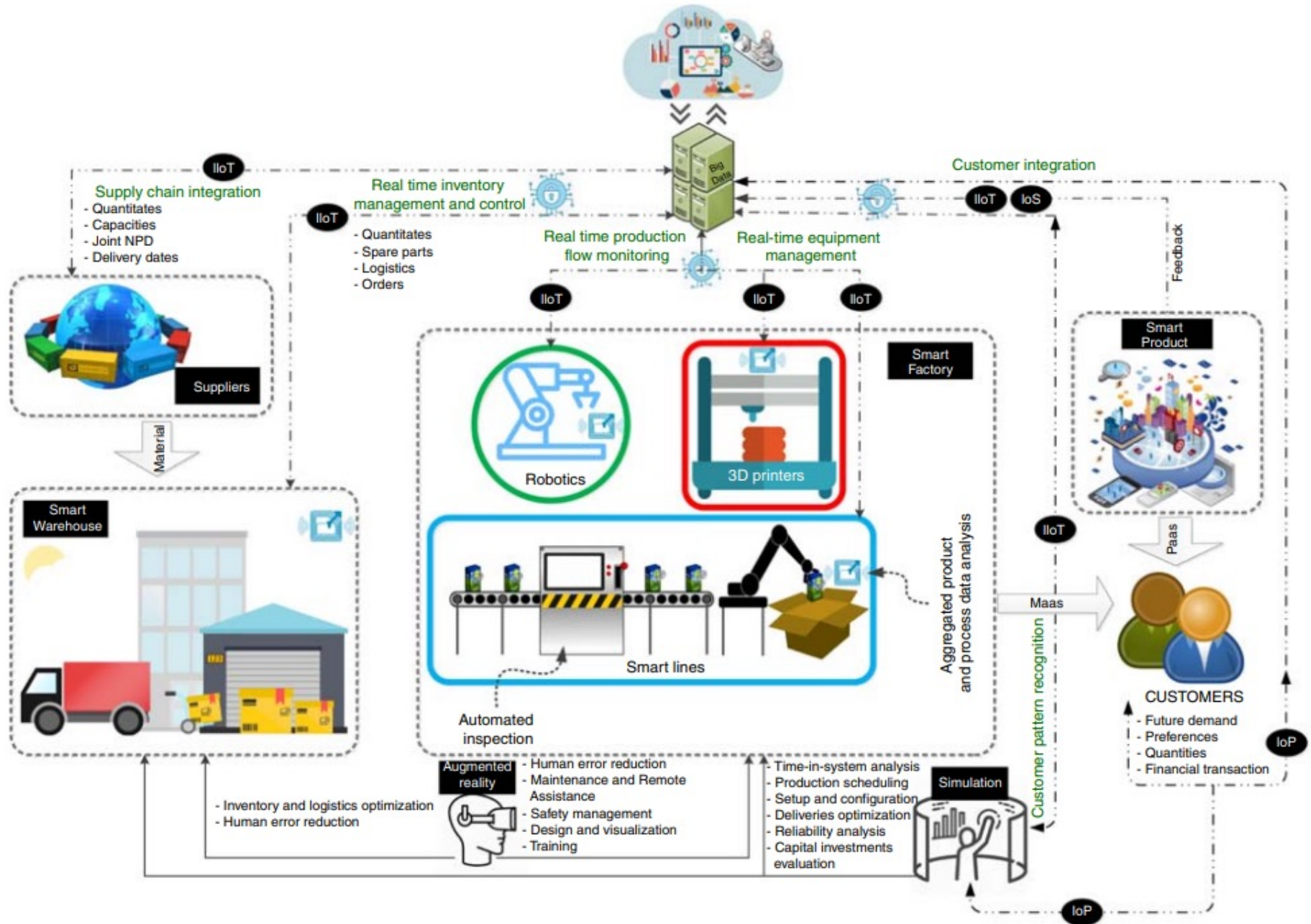
Cyber-Physical System

A Cyber-Physical System (CPS) is a system that integrates physical and computational components to monitor and control the physical processes seamlessly.

In other words, A cyber-physical system is a collection of computing devices communicating with one another and interacting with the physical world via sensors and actuators in a feedback loop.



Artificial Intelligence (AI) and Machine Learning (ML)



Challenges of Mechatronic Systems

- Design
- Privacy and Security
- Complexity and Ethics
- Ageing Population
- Sustainability
- Education



Privacy and Security

- Many of the devices associated with the IoT have the capacity to gather large volumes of personal data.
- This data is then subject to the possibility of analysis, with associated risks of misinterpretation impacting on privacy there is an increasing burden on system designers to place privacy at the core of their design process within the context each of the Internet of Things, Cyber-Physical Systems and Big Data, and that this must be reflected in the design process itself and the methods and tools to support this.
- Information Systems Audit and Control Association (<https://www.isaca.org/>)

Threat	Probability (%)
Data leakage	17
Employee error	16
Employee-owned device incidents	13
Cloud computing	11
Cyber attacks	7
Disgruntled employee	5
External hacking	5
All of the above	19
None of the above	8

Perceived threats to system security

Complexity and Ethics

- Dual-use of technology
- Impact of a technology on the environment
- Impact of technology on the global distribution of wealth
- The digital divide and the associated socio-technological gap
- Ensuring fair access to technologies
- Technological addiction
- Technological lock-in
- Dehumanisation of humans and anthropomorphism



Military drone



Agriculture drone

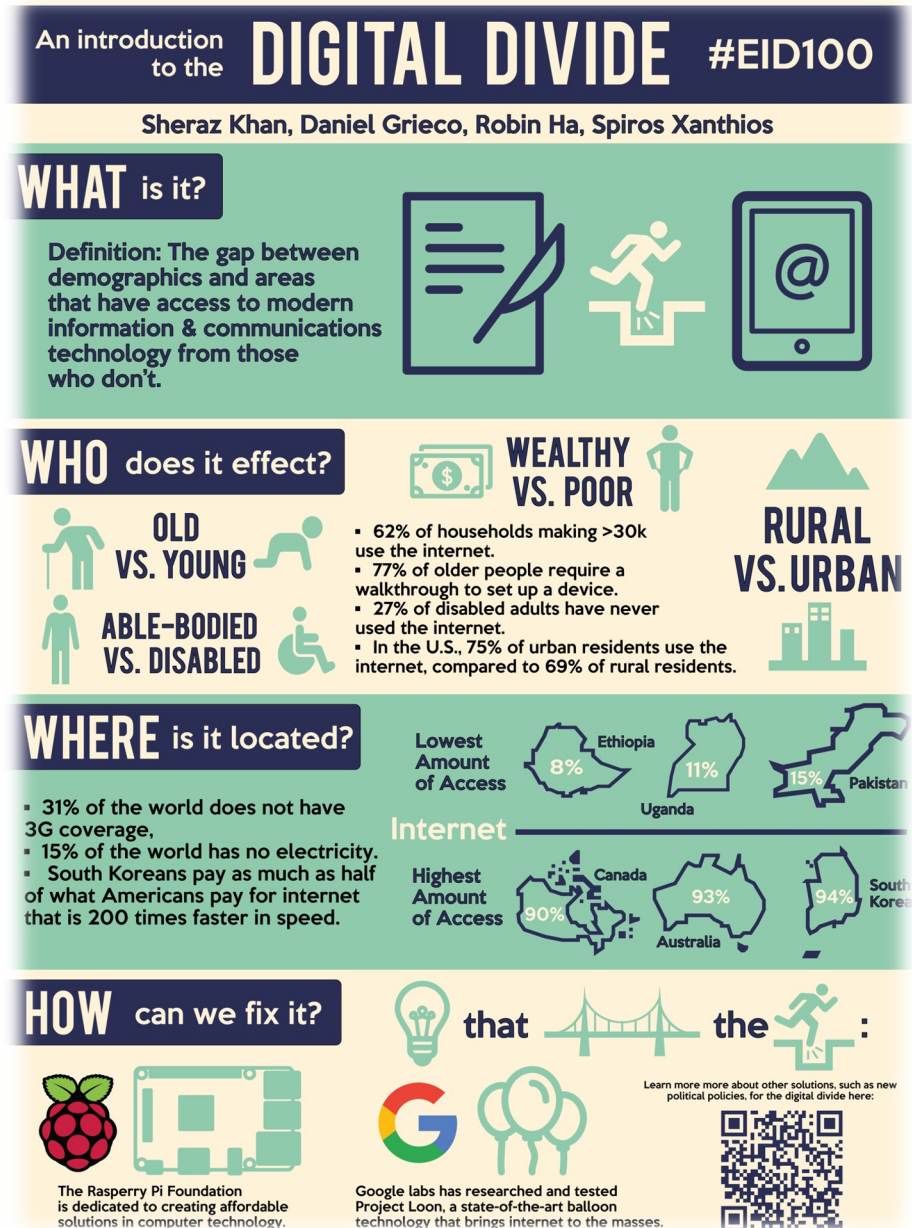
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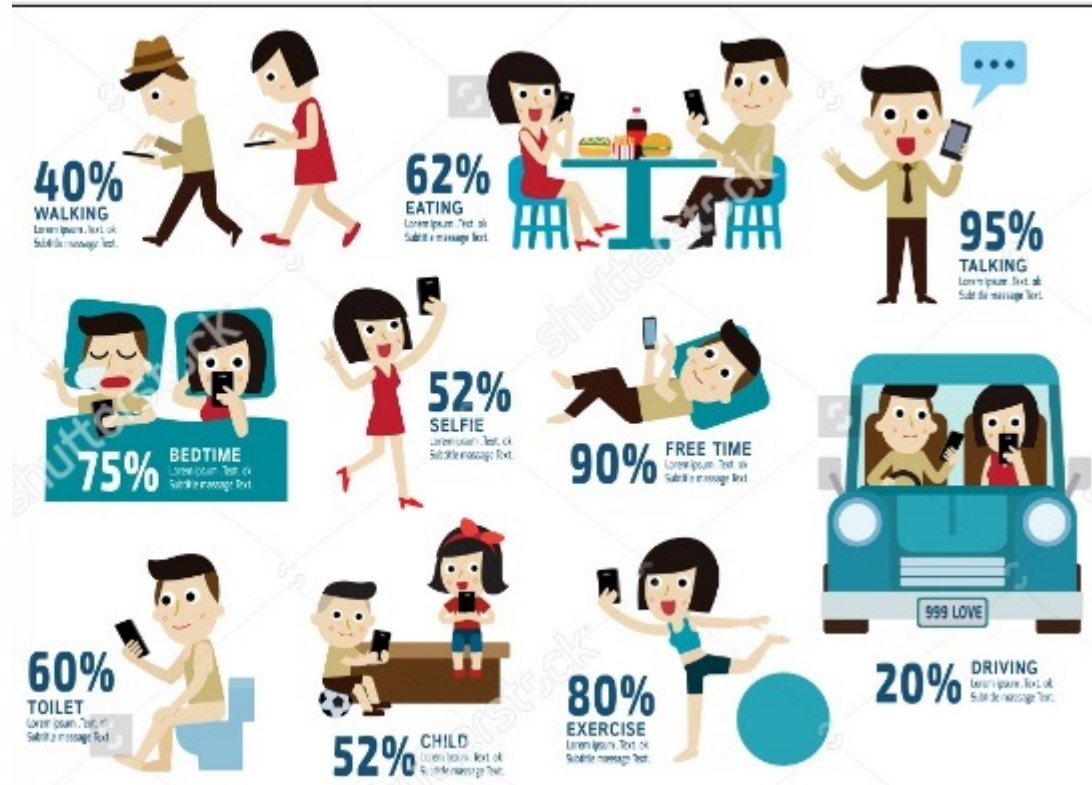
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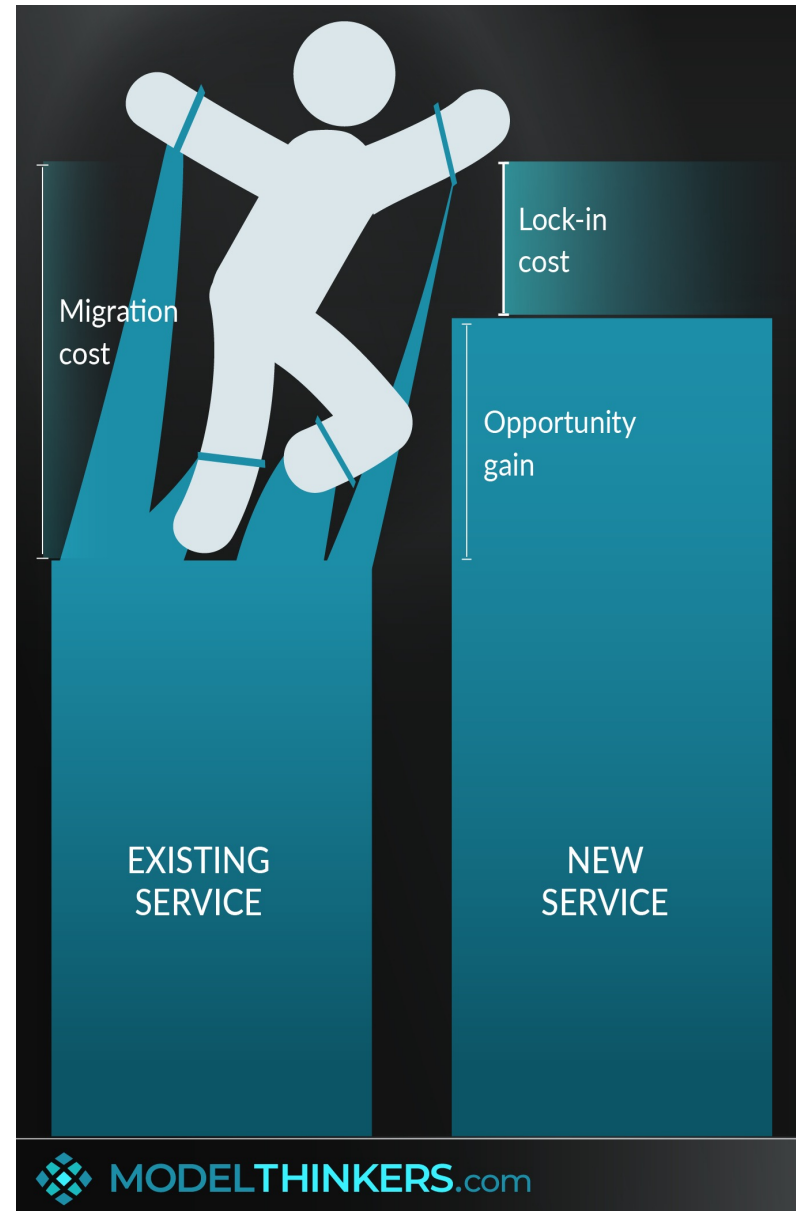
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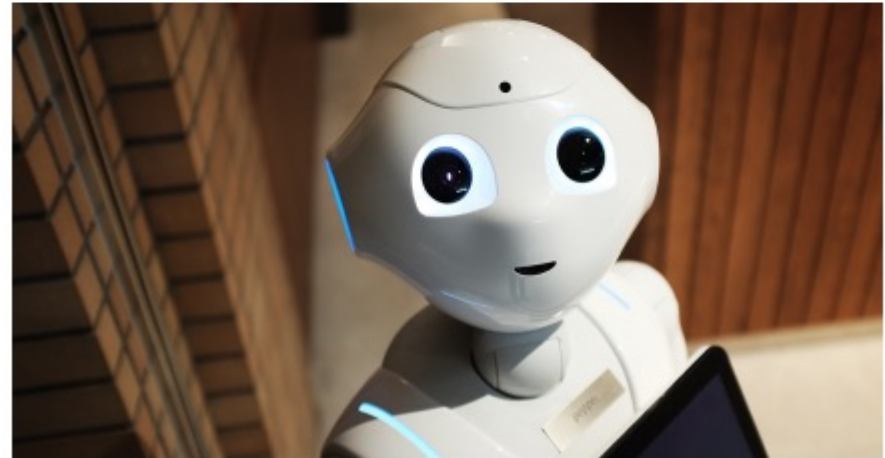
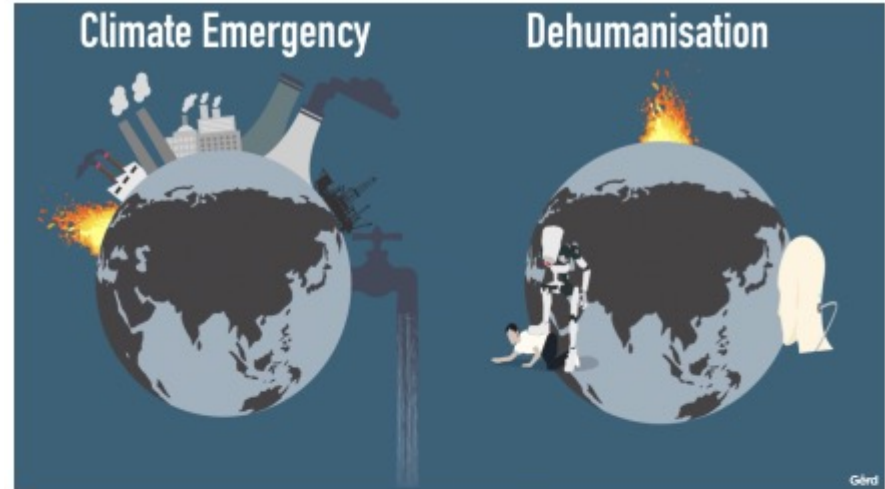
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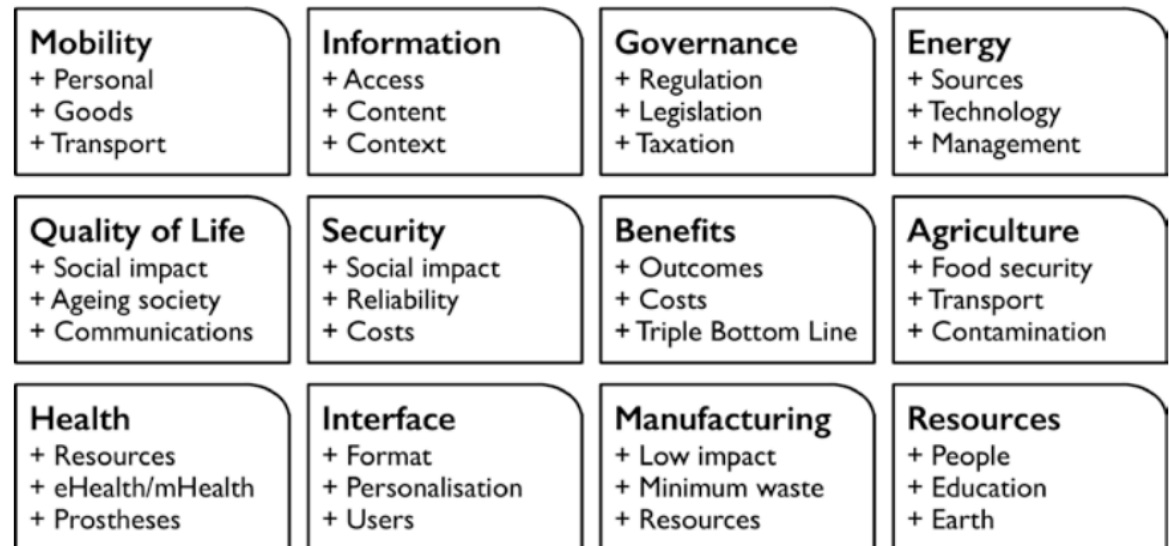
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Sustainability

- A recognised need to move towards more sustainable forms of society-centred around the individual and their needs and structured around the effective management and use of all available resources.
- Concepts such as those of the smart home and the smart city where information is used to manage daily activities
- Creation of sustainable societies with individuals and the core to address issues such as ageing populations, resource availability and management, climate change and resilience



	US/Canada	Europe	Asia	Latin America	Africa
Average population (millions)	1.4	2.5	9.4	4.6	3.9
Population density (per km ²)	3100	3900	8200	4500	4600
Water consumption (litres per capita per day)	587	288	278	264	187
Water loss rate (%)	13	23	22	35	30
CO ₂ emissions per capita (tonnes)	14.5	5.2	4.6	No data	No data
Waste volume (kg per capita per year)	No data	511	375	465	408

Education

Mechatronics education has always faced the challenge of balancing appropriate levels of technical content with the understanding of the requirements for integration across the core disciplines of mechanical engineering, electronics and information technology.

- **Changes in delivery**

- Massive On-line Open Courses (MOOCS).
- Tutorial and workshop based learning support.
- Blended learning
- Impact of social media on learning.

- **Structural Issues**

- Distributed learning resources.
- Time value of content.
- Collaborative working

